

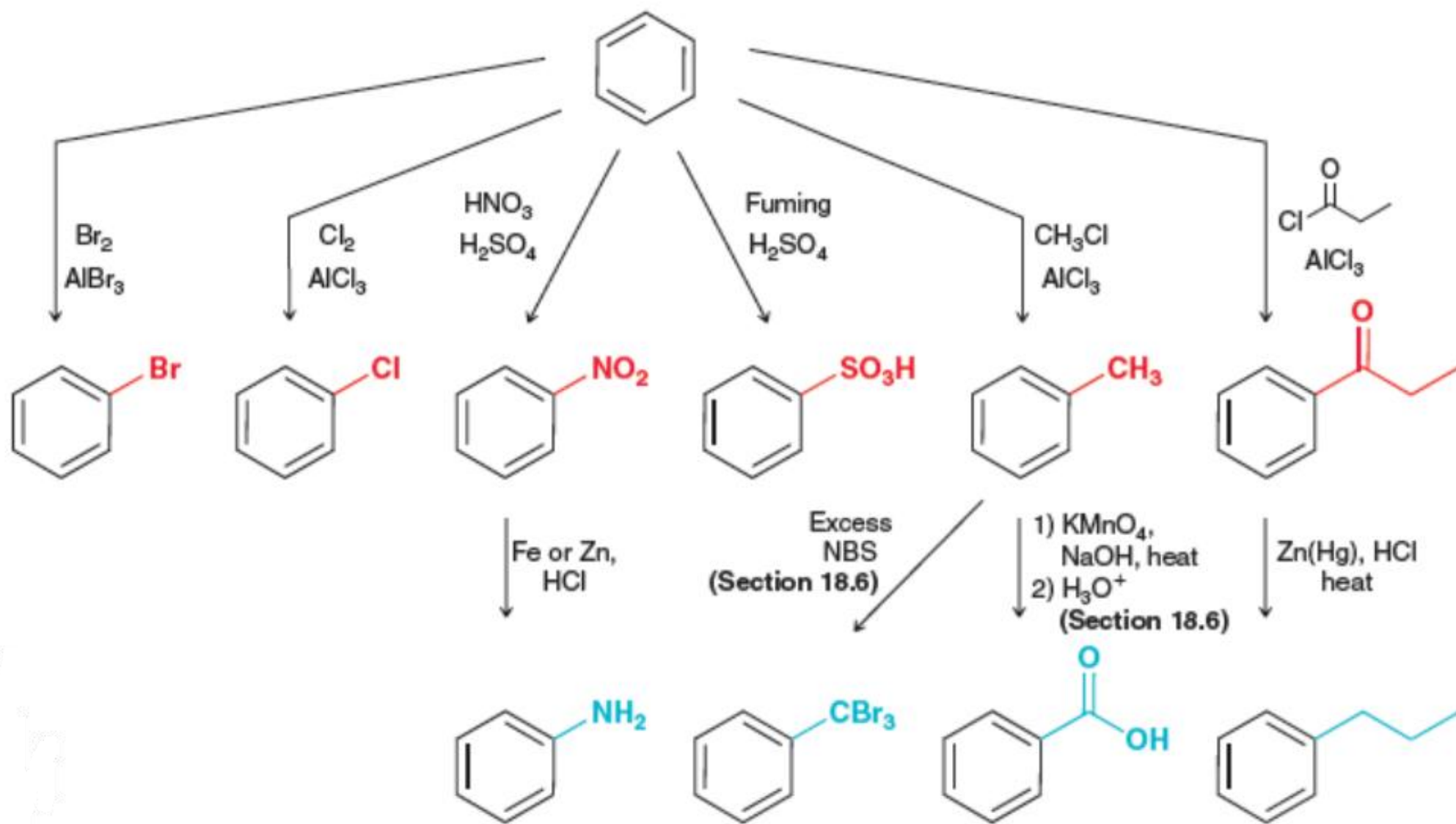
有机化学B专题讲座1

芳香亲电取代反应合成策略

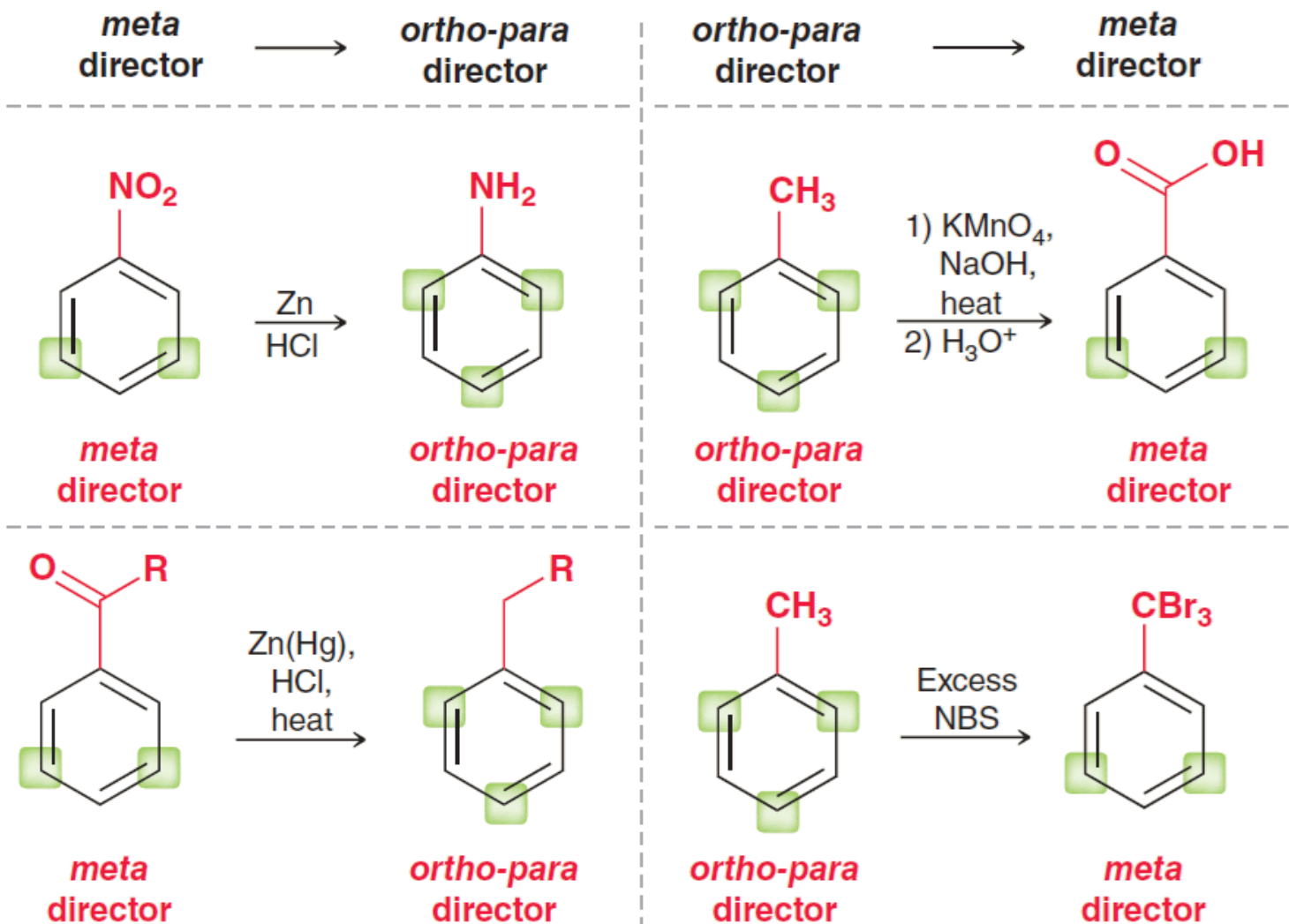
佟 硕

2025年4月19日

芳香亲电取代反应

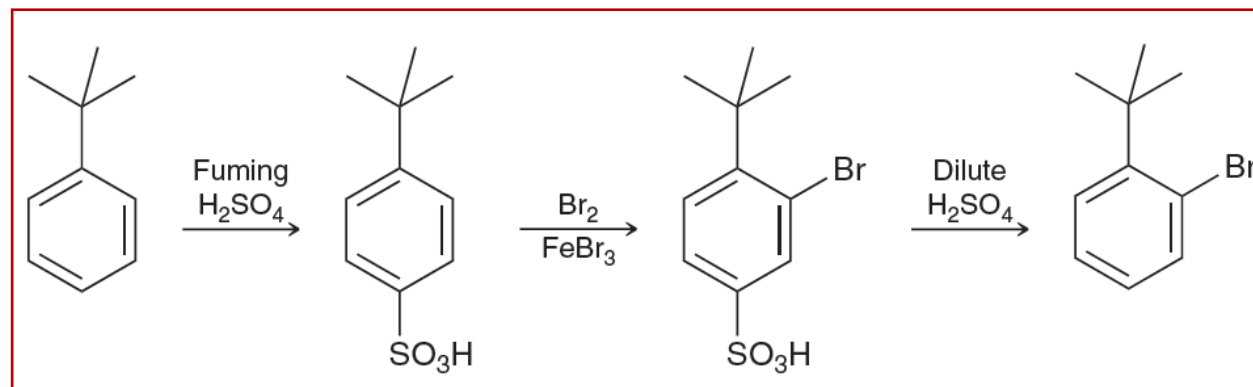
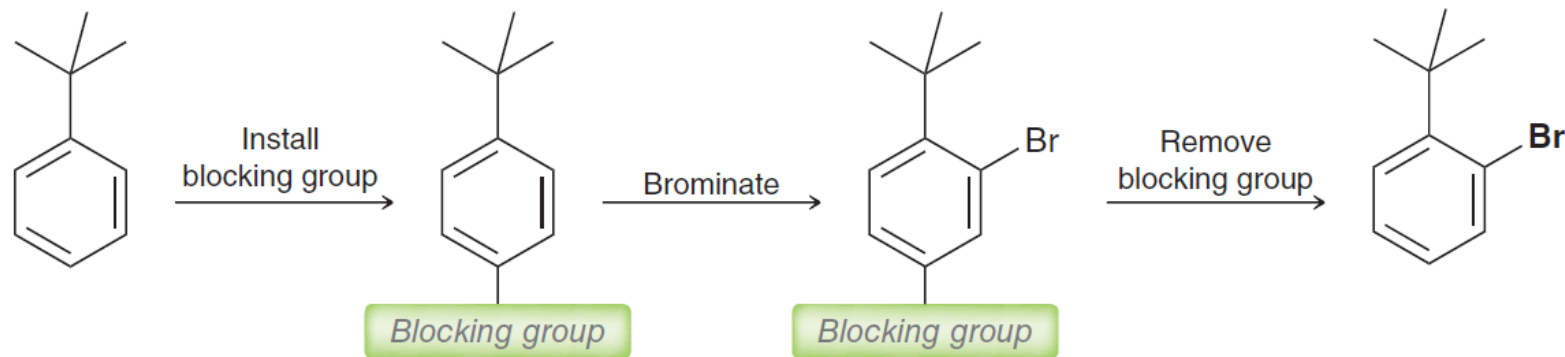
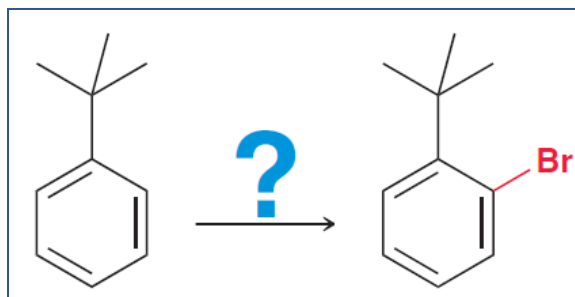
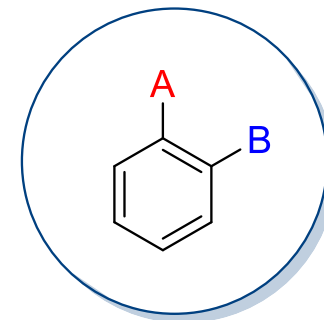


定位规律的转换



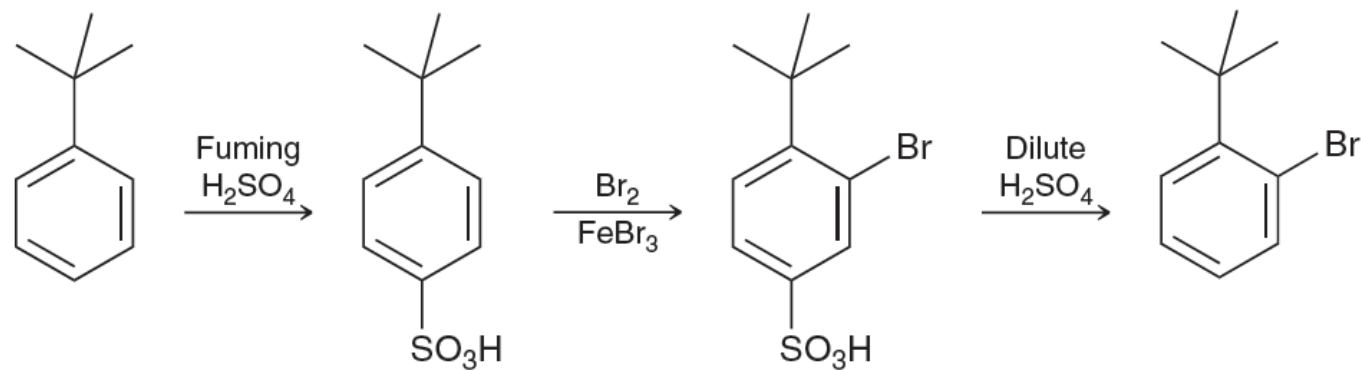
芳香亲电取代反应的合成策略

- 利用磺化反应临时占位，生成邻位产物



何为好的合成策略？

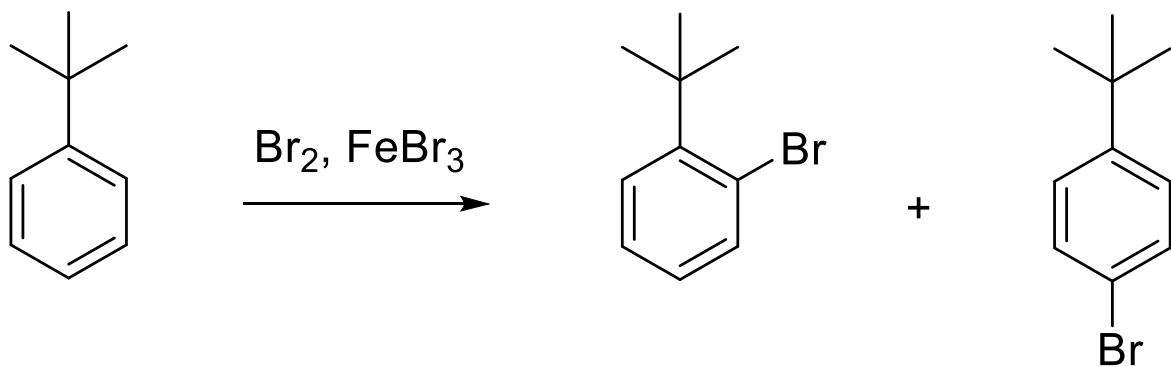
Route 1



优

VS

Route 2



副产物

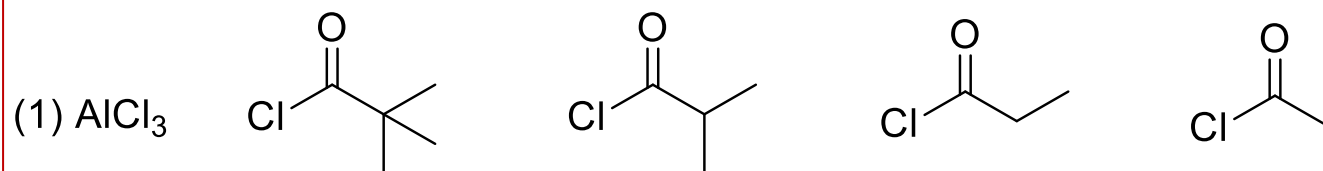
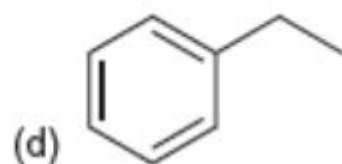
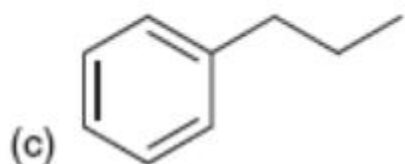
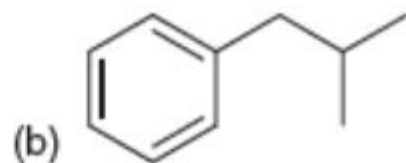
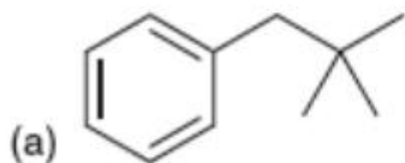
主产物

劣

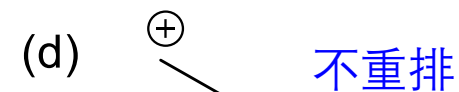
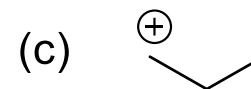
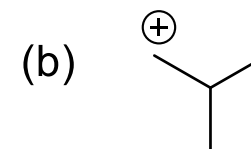
芳香亲电取代反应的合成策略

- 付-克烷基化反应中的碳正离子重排以及多取代问题

19.8 Identify whether each of the following compounds can be made using a direct Friedel-Crafts alkylation or whether it is necessary to perform an acylation followed by a Clemmensen reduction to avoid carbocation rearrangements:

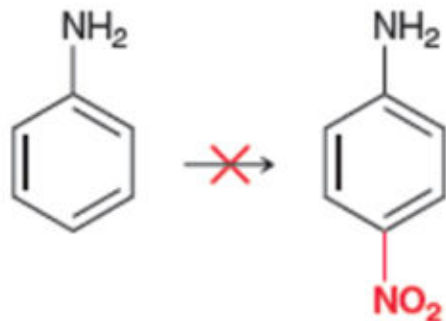


(2) Zn/Hg HCl (conc.)



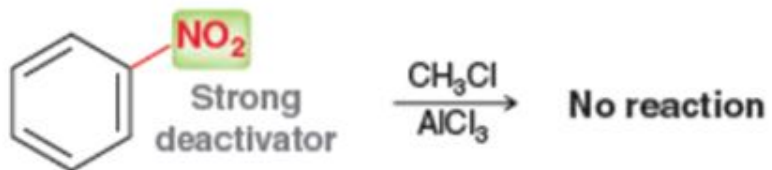
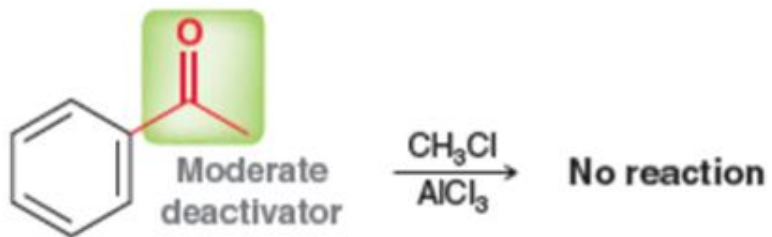
芳香亲电取代反应的合成策略

- 氨基与硝化条件不兼容



HNO₃/浓H₂SO₄为氧化条件
NH₂官能团不耐受

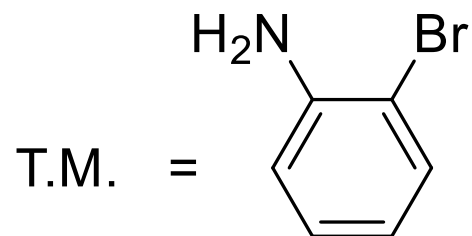
- 含有中等/强钝化基团的芳环不能发生付-克烷基化和付-可酰基化反应



E⁺活性有别

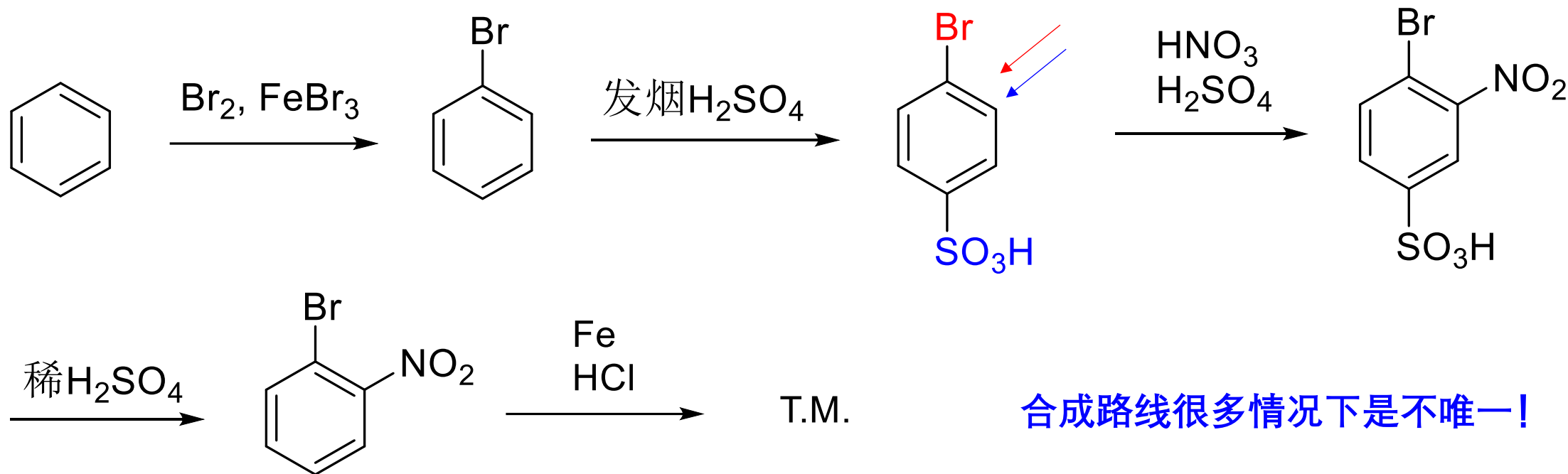
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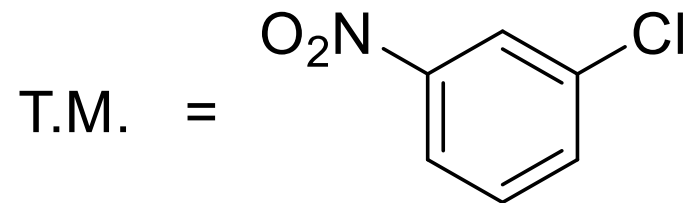
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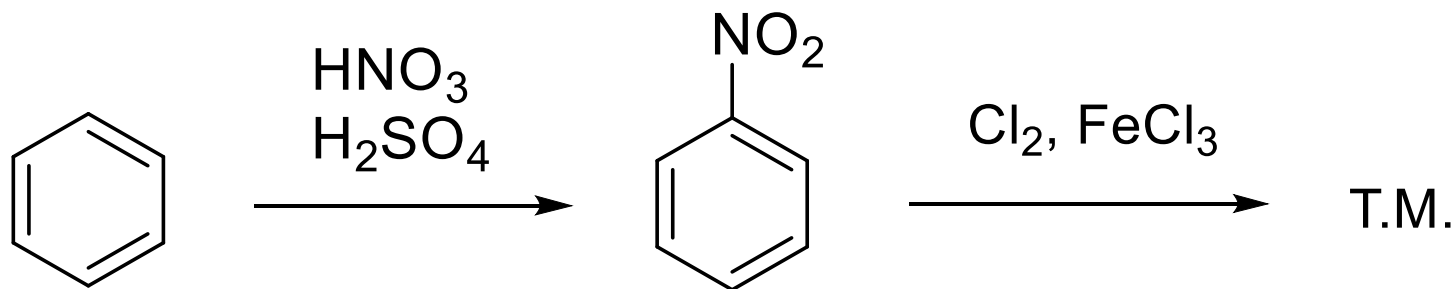
目标分子
Target molecule
T.M.

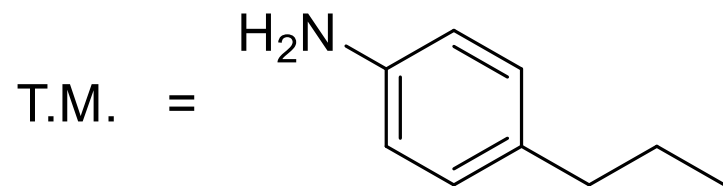
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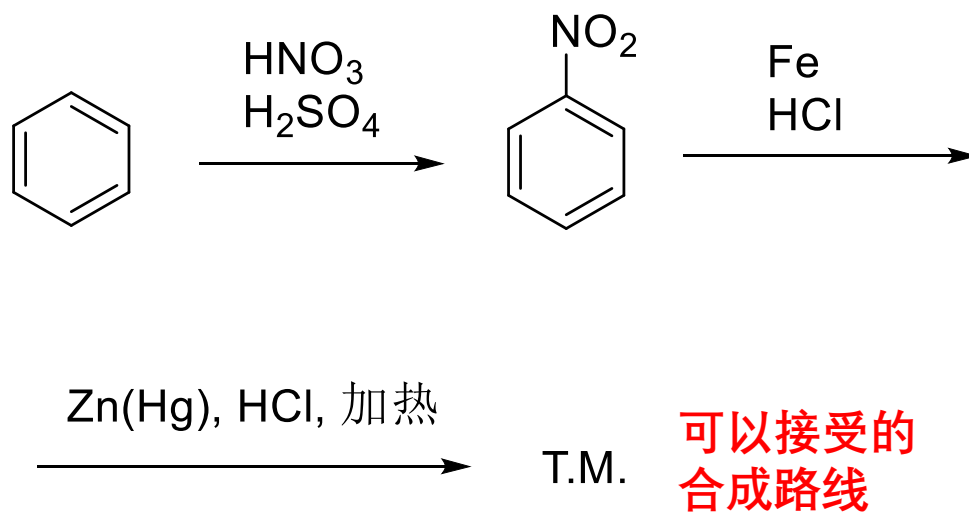
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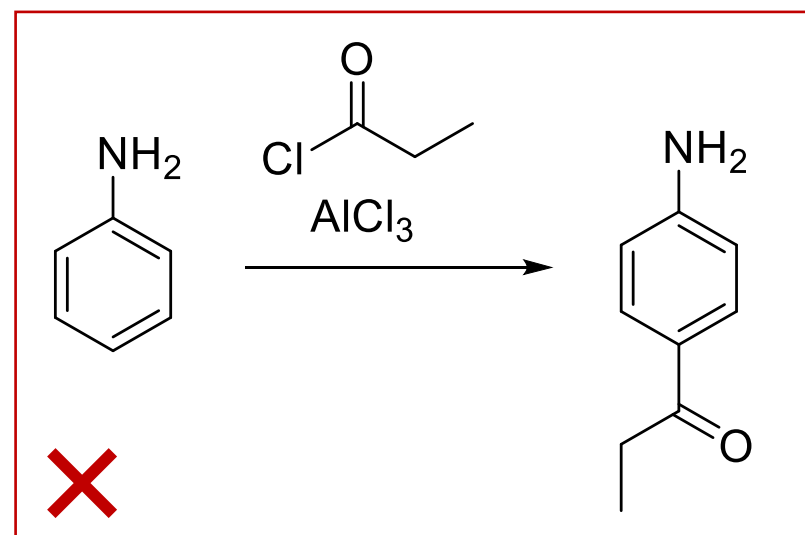


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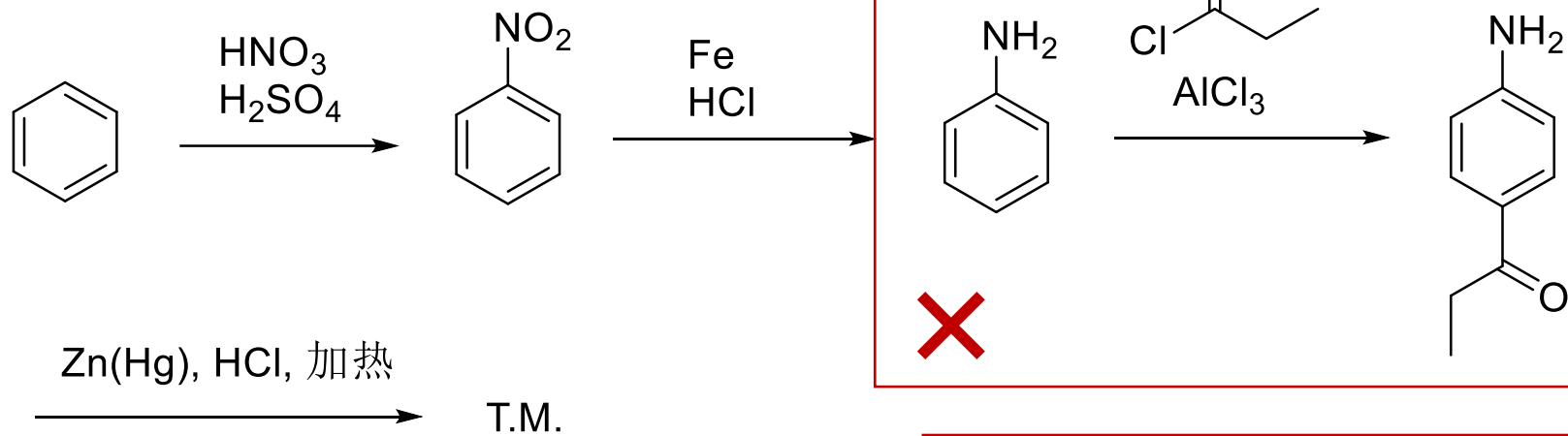
Route 1



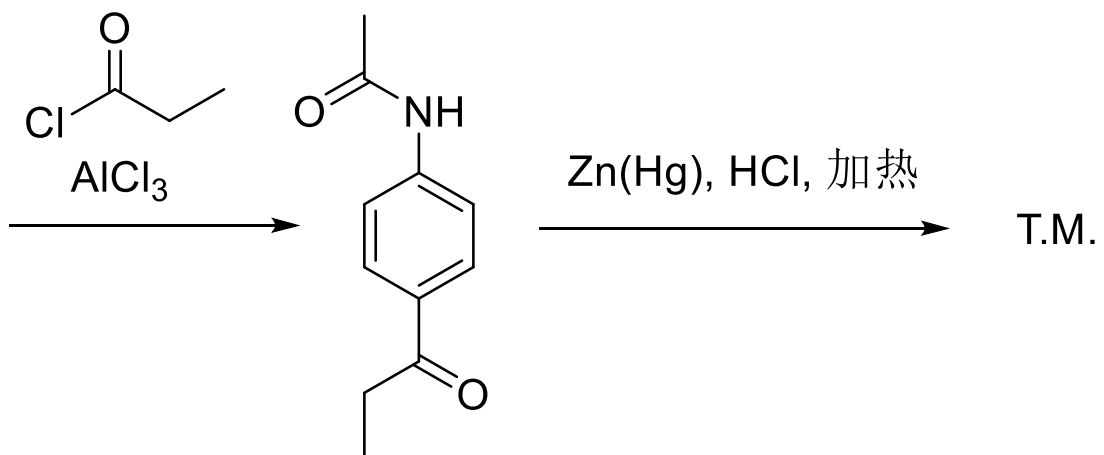
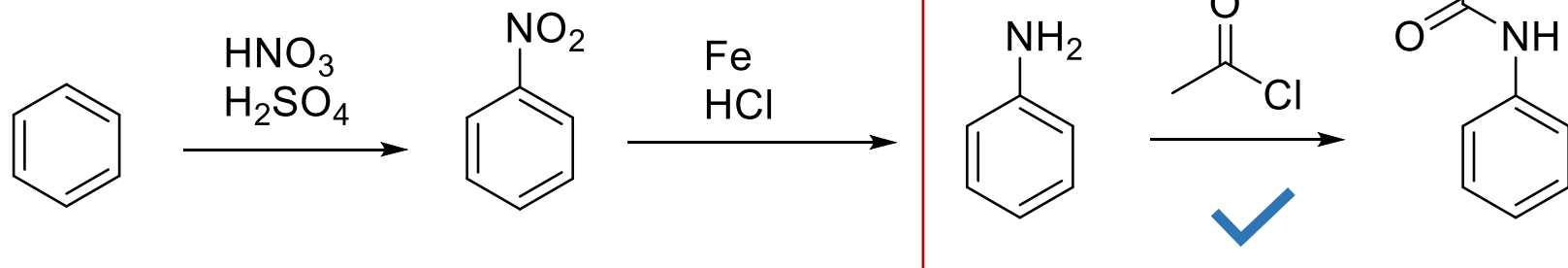
可以接受的
合成路线



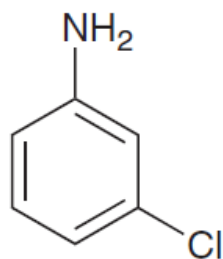
Route 1



Route 2

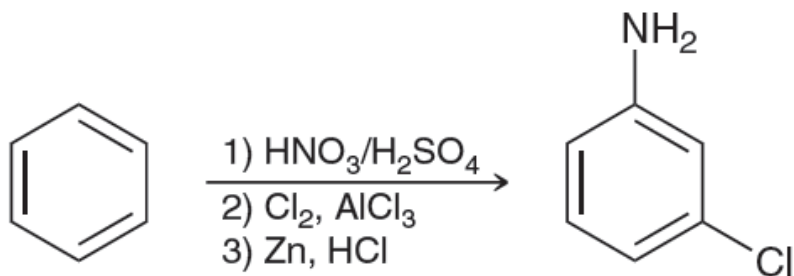
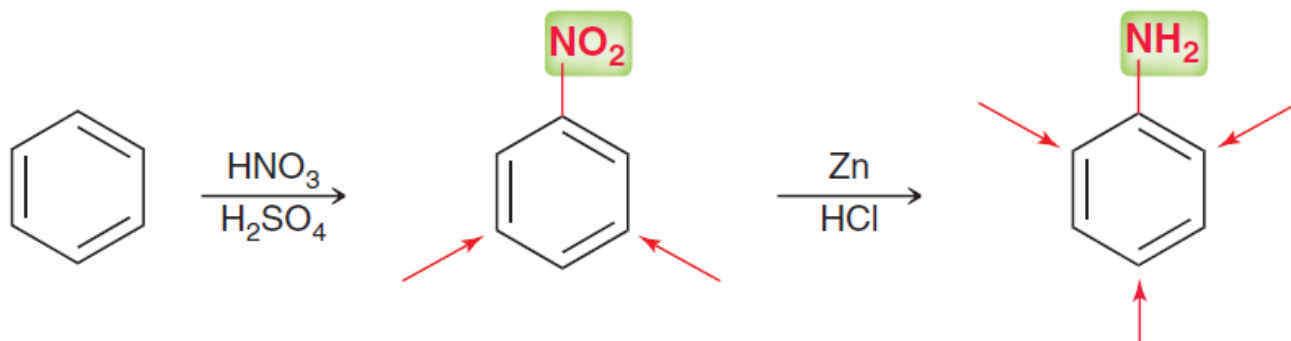


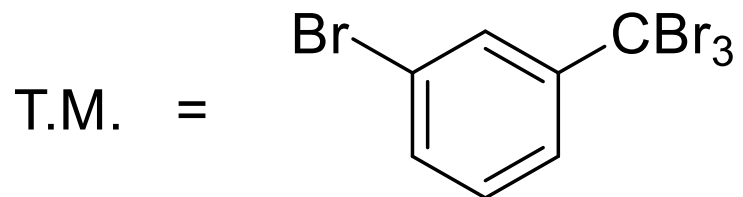
合成路线不唯一，但有好坏之分！



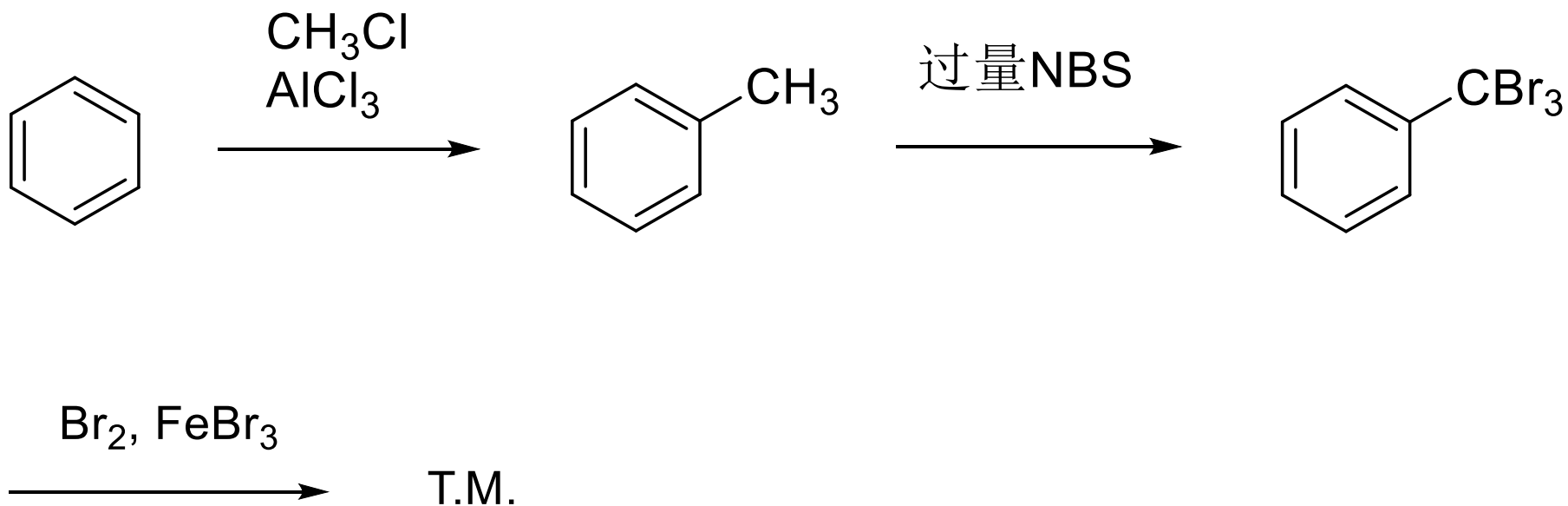
目标分子 Target molecule T.M.

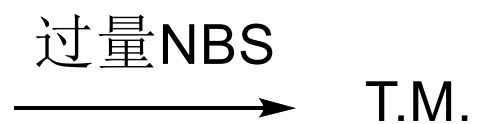
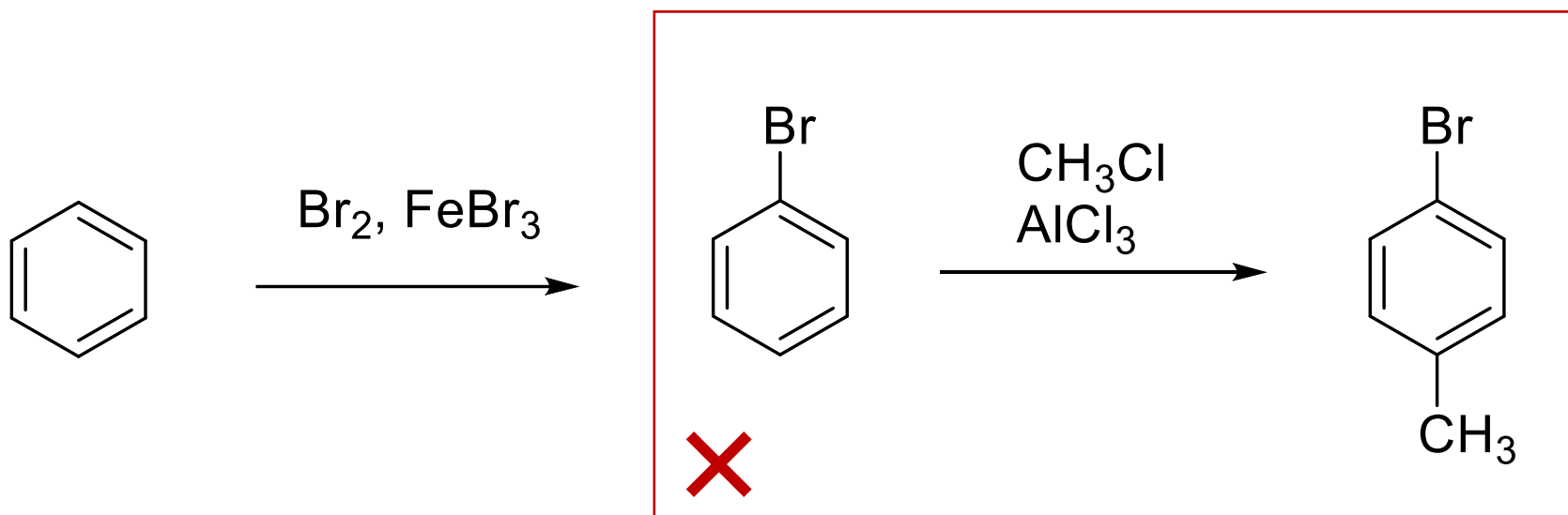
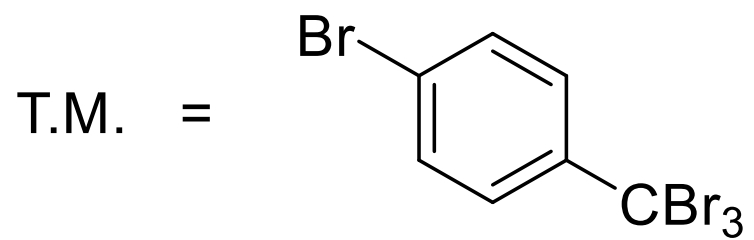
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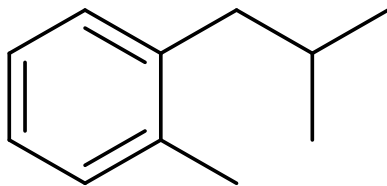
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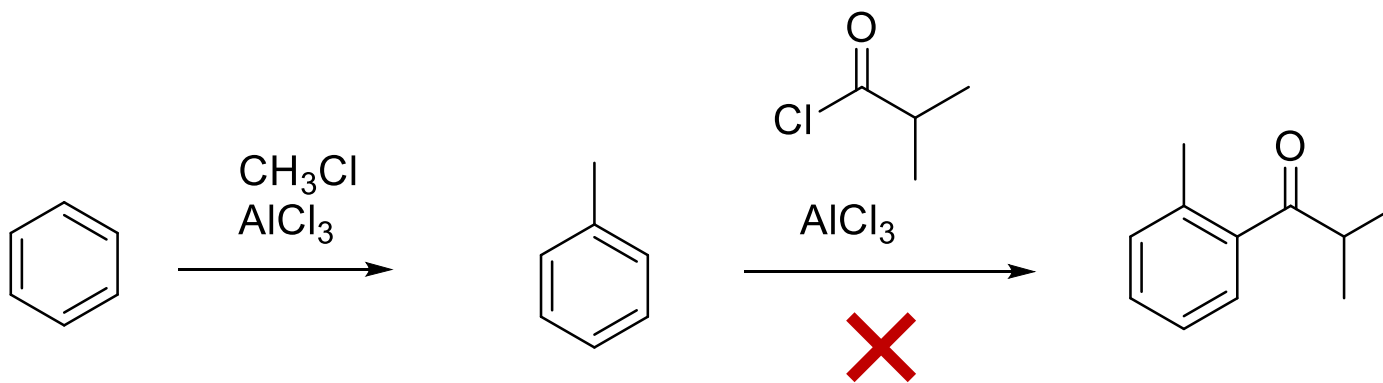
合成路线并不总能尽善尽美!

T.M. =



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Route 1

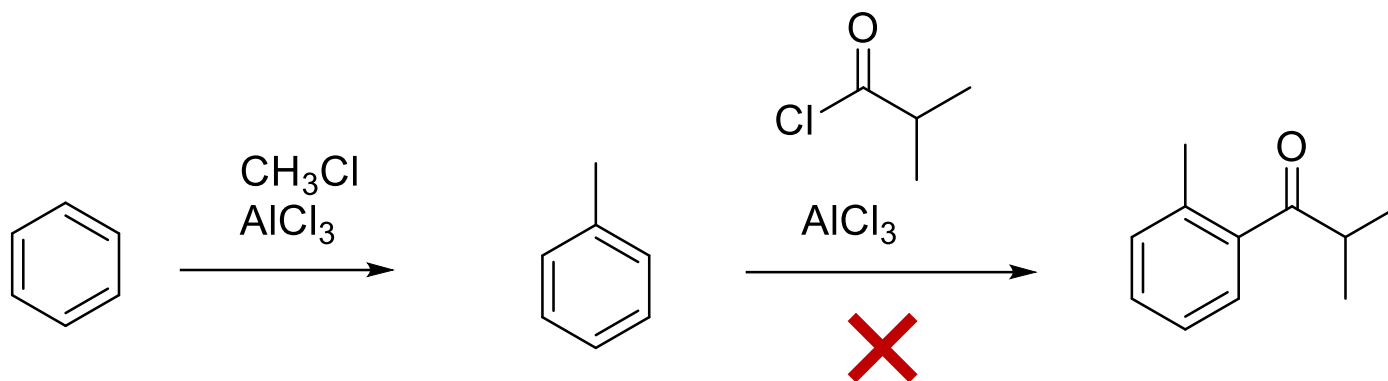


Zn(Hg), HCl, 加热



T.M.

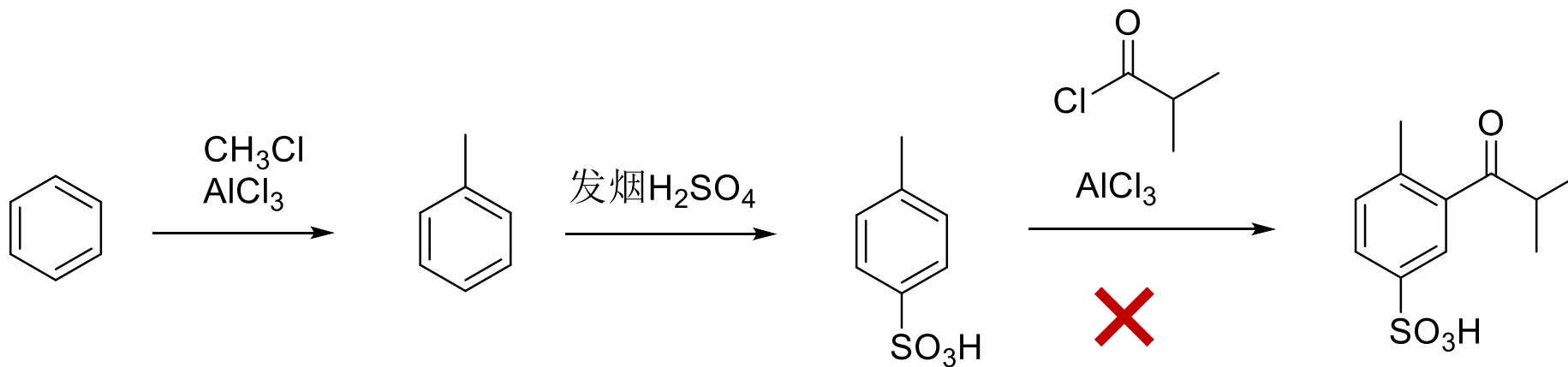
Route 1



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T.M.

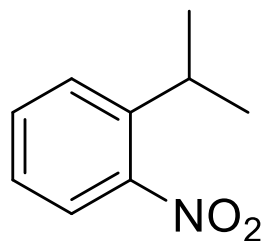
Route 2



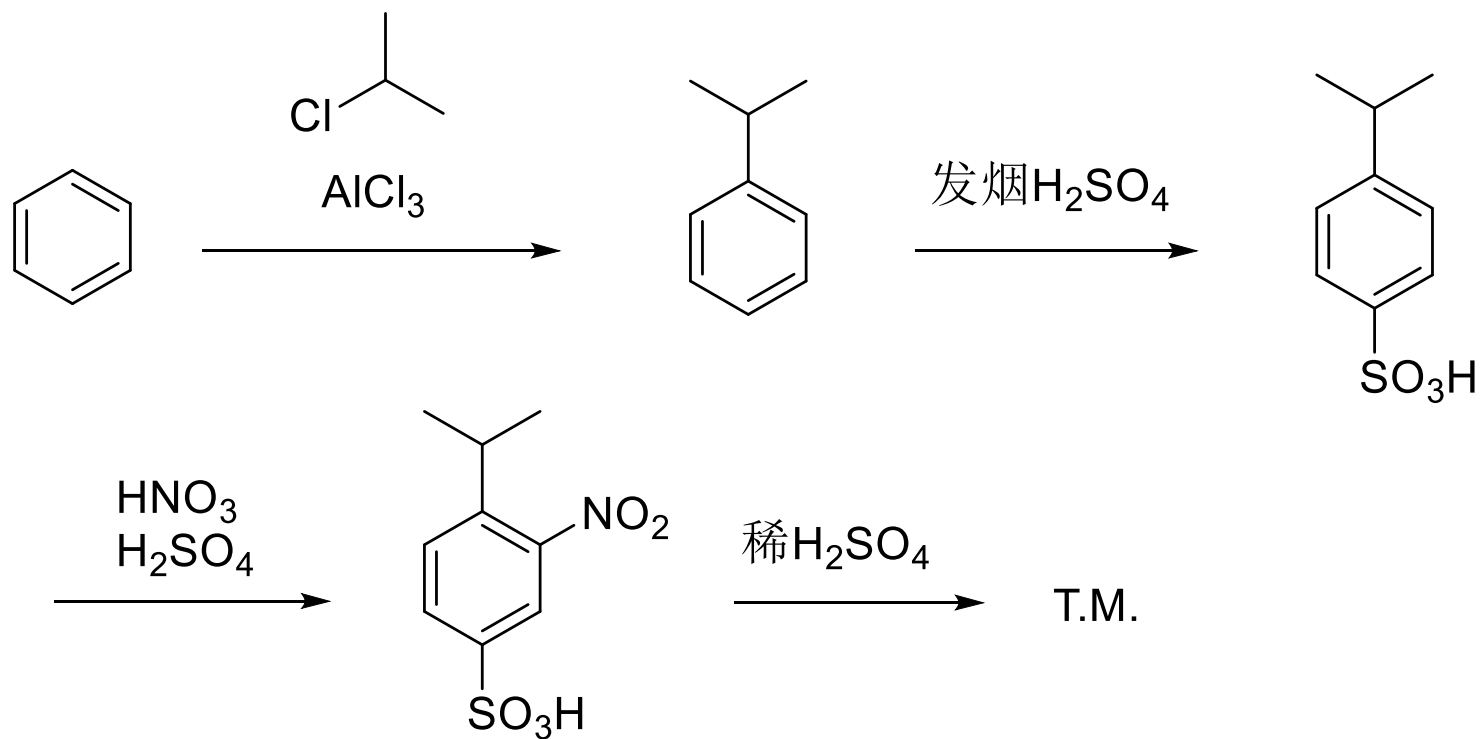
Zn(Hg) , HCl , 加热

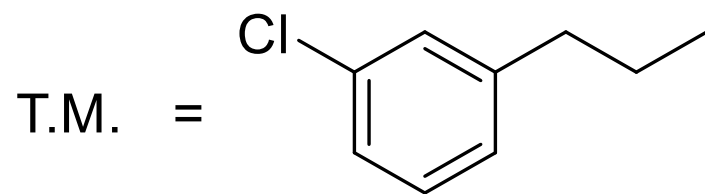
T.M.

T.M. =

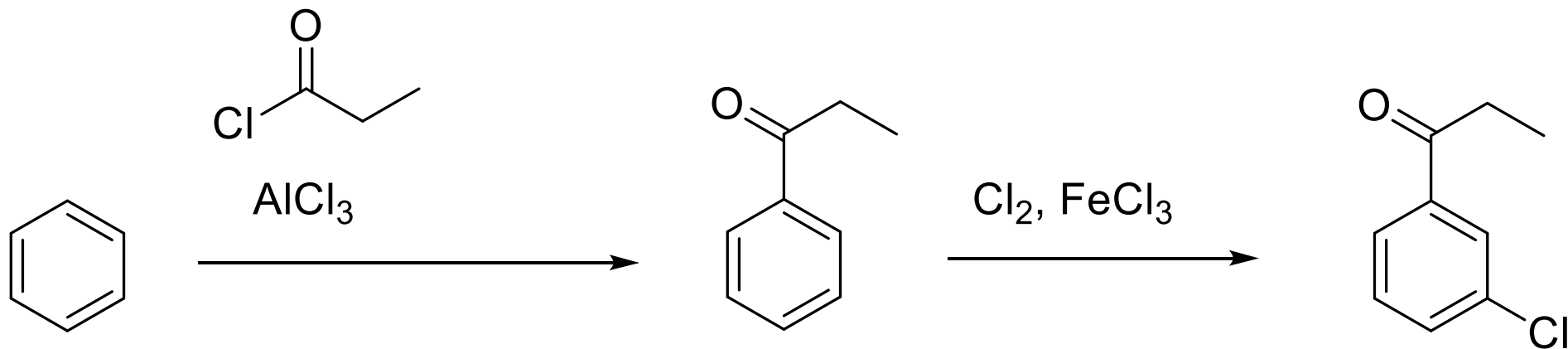


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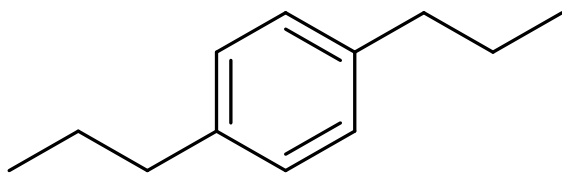


Zn(Hg), HCl, 加热

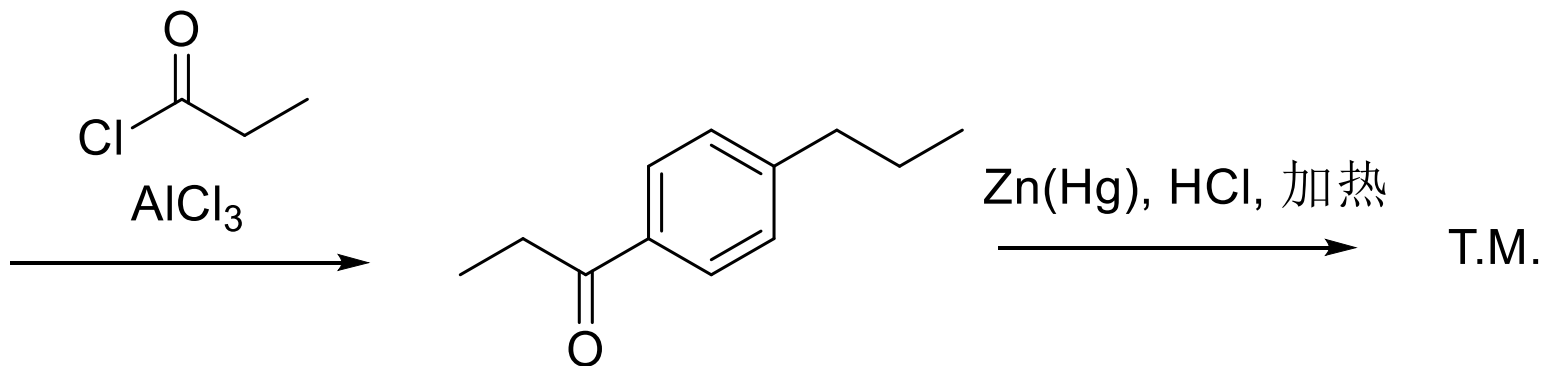
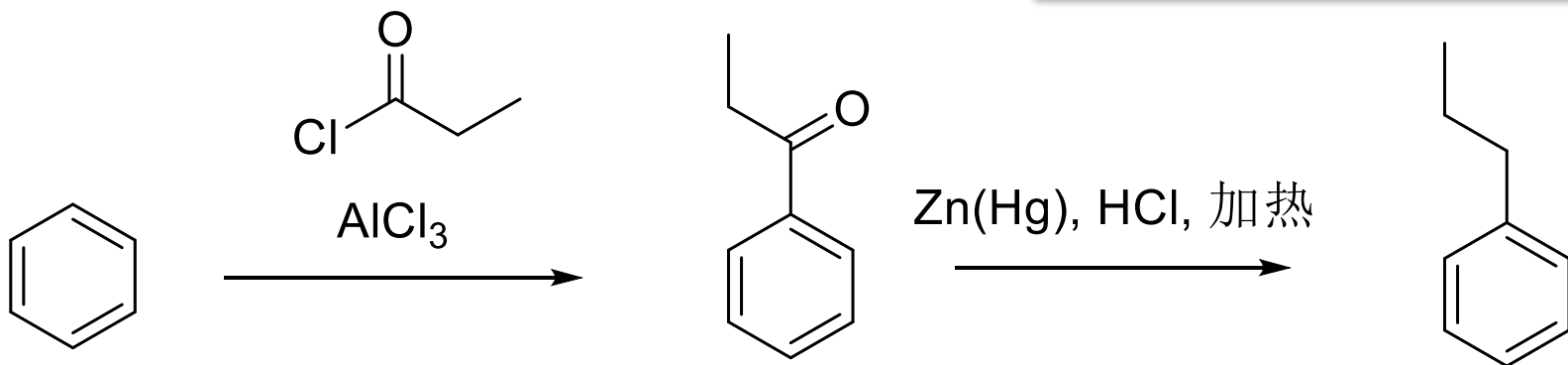


T.M.

T.M. =

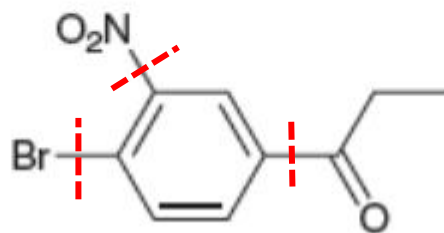


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更复杂的多步合成问题

Starting with benzene and using any other necessary reagents of your choice, design a synthesis for the following compound:



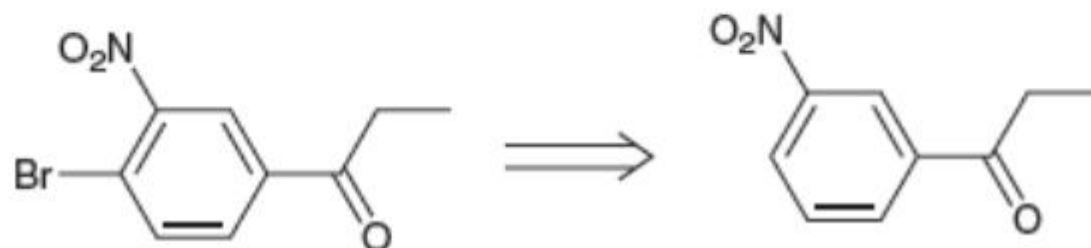
逆合成分析



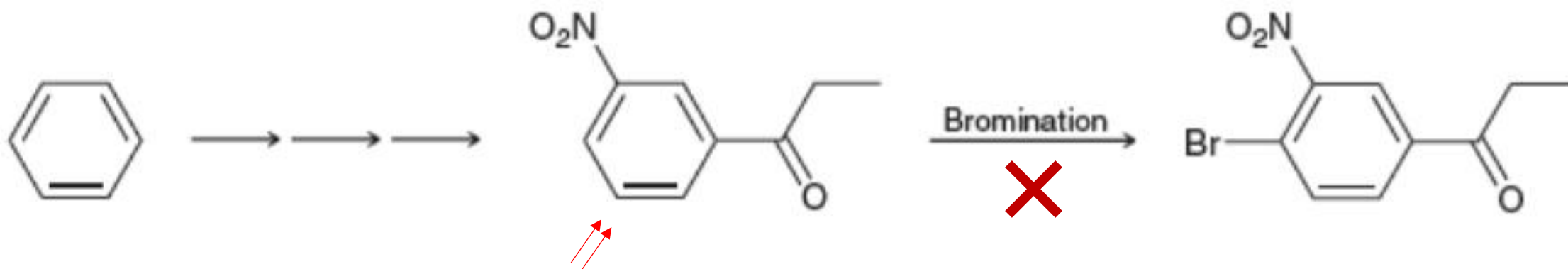
SOLUTION

Approaching this problem from a retrosynthetic point of view, we begin by determining the last step of the synthesis. There are three possibilities: (1) the Br group is installed last, (2) the NO₂ group is installed last, or (3) the acyl group is installed last.

Let's begin by supposing that the Br group is installed last.

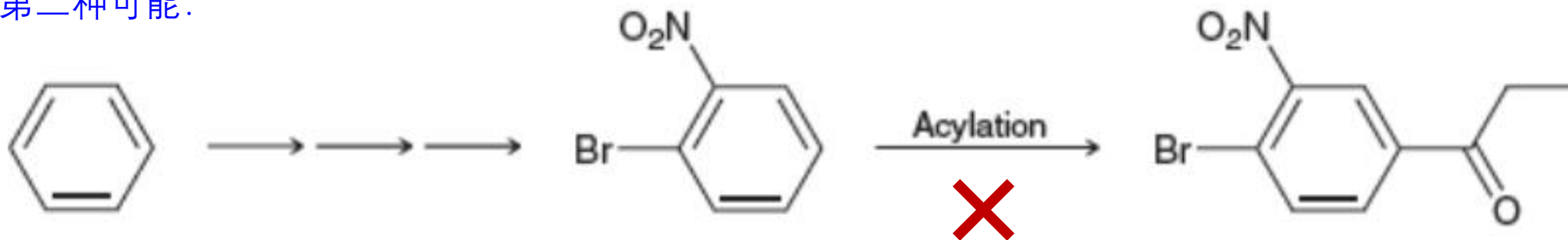


第一种可能：



To consider the plausibility of this as our last step, we must first examine whether the desired regiochemical outcome will be achieved. In this case, we are trying to achieve the bromination of a disubstituted ring in which both groups (the nitro group and the acyl group) are *meta* directors. In such a case, the incoming Br group would be installed *meta* to both groups, which is not the desired location. Therefore, this step cannot be the last step of our synthesis.

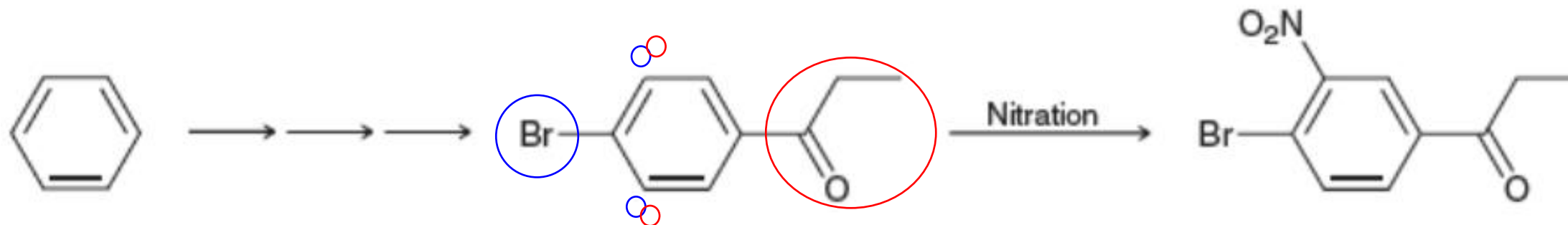
第二种可能：



In this case, the desired regiochemical outcome would be achieved because both the Br group and the NO₂ group direct to the desired location. However, this transformation is a Friedel-Crafts acylation, so we must consider whether the proposed reaction violates any of the limitations for Friedel-Crafts acylation. In fact, there is a violation here because the ring is strongly deactivated by the presence of a nitro group, and the desired reaction simply cannot be achieved.

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第三种可能：

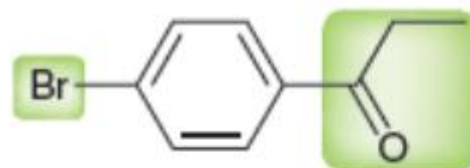


Both the Br group and the acyl group will direct the incoming nitro group to the desired location, and this reaction is plausible. This must be the last step.

STEP 2

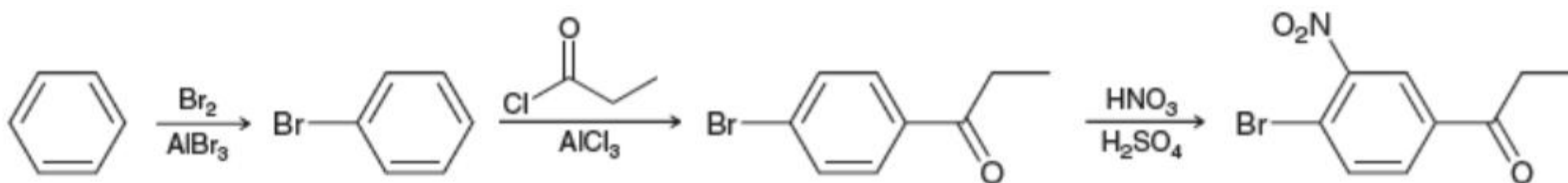
Determine the penultimate (second-to-last) step of the synthesis.

Continuing to work backward, we must now consider the order in which the remaining two groups should be installed:

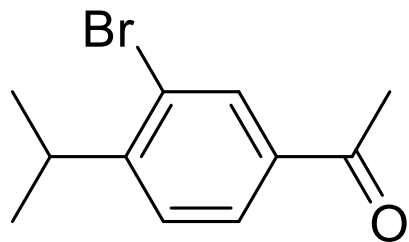
**STEP 3**

Consider any limitations for a proposed step.

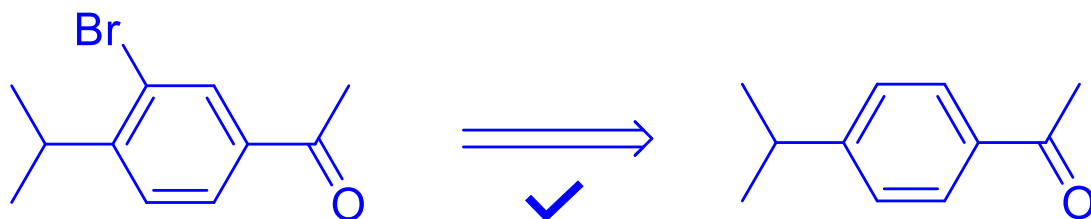
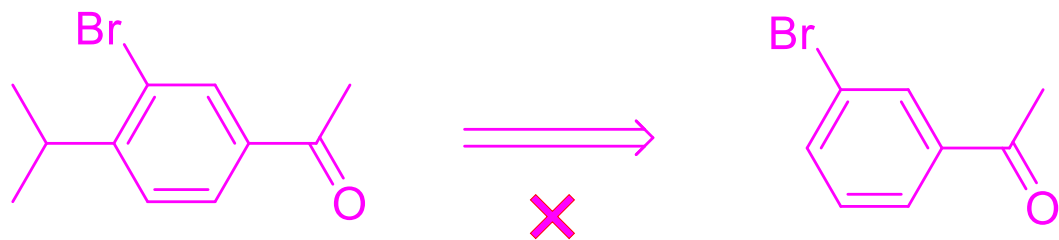
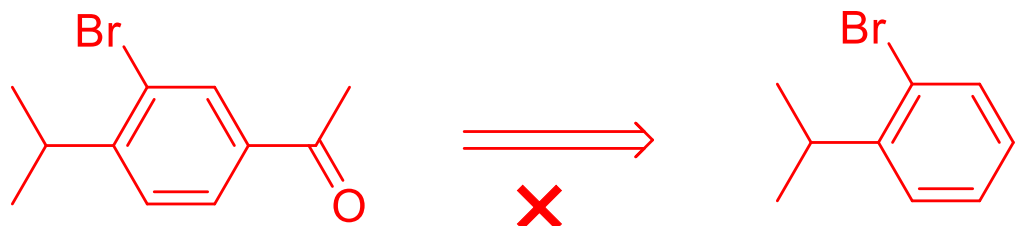
Once again, we must choose a sequence of events that achieves the desired regiochemical outcome. The Br and acyl groups are *para* to each other, so we must consider which group is a *para* director. Indeed, the Br group is an *ortho-para* director (with a preference for *para*), while the acyl group is a *meta* director. Therefore, we conclude that bromination be carried out first followed by acylation. Whenever performing an acylation step, it is necessary to consider the limitations of acylation. This case requires the acylation of bromobenzene. The Br group is only weakly deactivating, which does not interfere with the acylation process (acylation is only unattainable with moderately deactivated or strongly deactivated rings). In summary, our proposed synthesis has the following sequence of events:



T.M. =



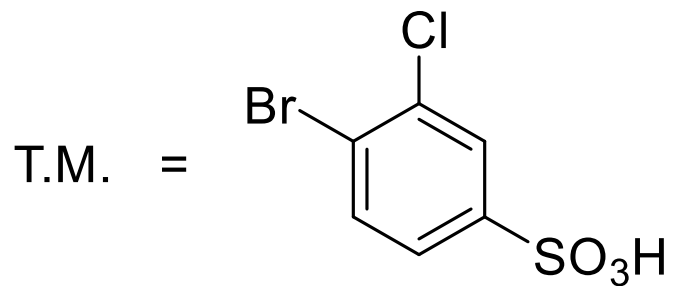
逆合成分析:



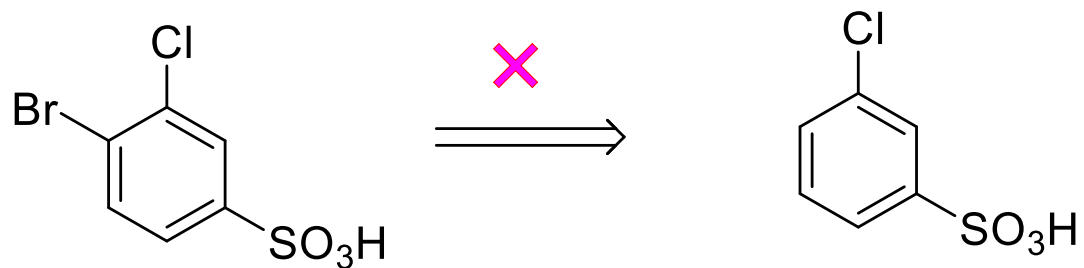
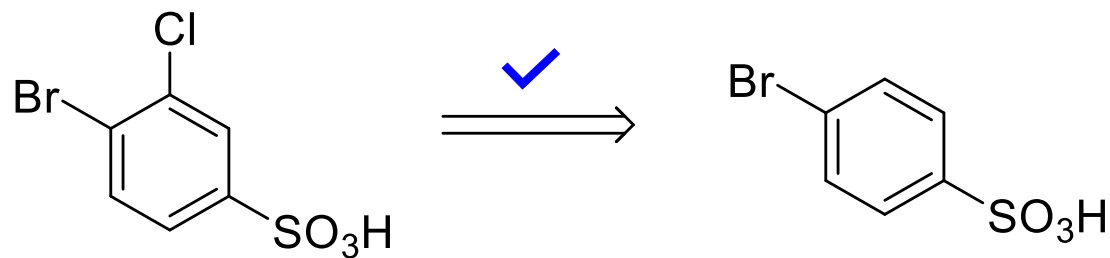
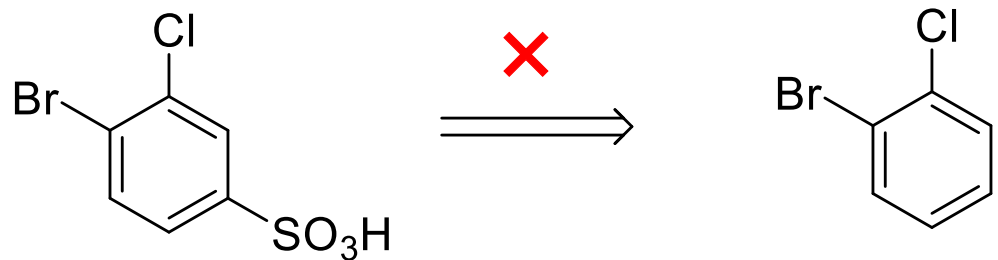
✗ 取代基的定位效应 (电子效应)

- 定位规律的转换
- 取代基的位阻效应
- 利用磺化反应临时占位, 生成邻位产物
- 付-克烷基化反应中的碳正离子重排以及多取代问题
- 氨基与硝化条件不兼容

✗ 含有中等/强钝化基团的芳环不能发生付-克烷基化和付-可酰基化反应

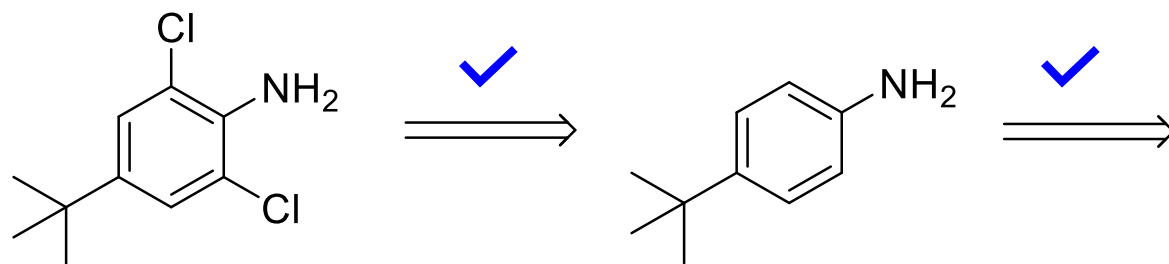
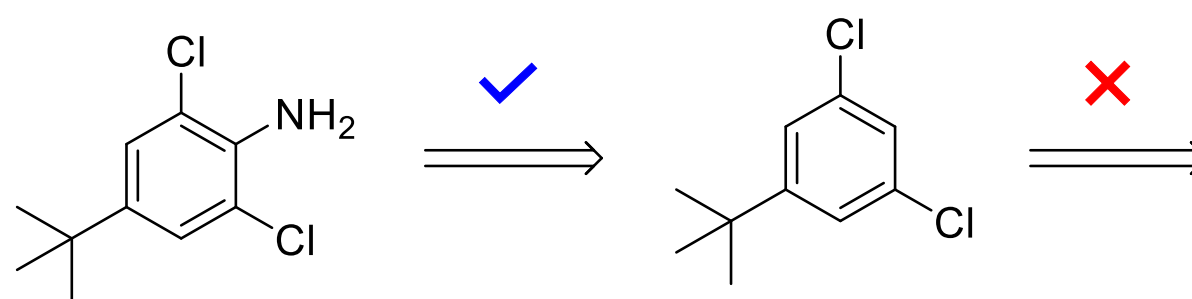
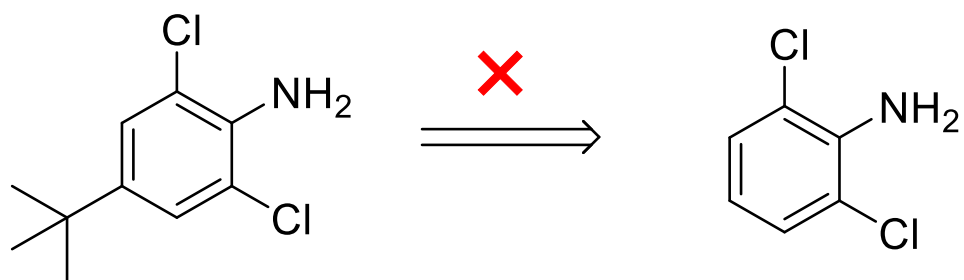
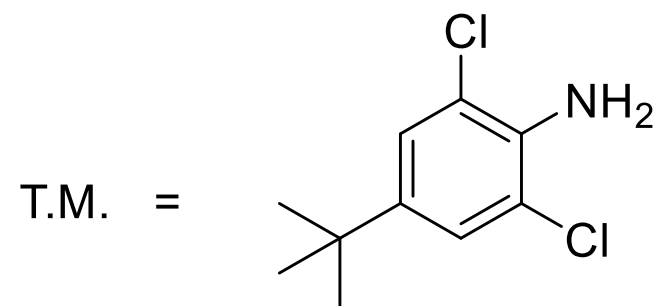


逆合成分析:



✗✗ 取代基的定位效应 (电子效应)

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芳香亲电取代反应的合成策略

- 取代基的定位效应（电子效应）
- 定位规律的转换
- 取代基的位阻效应
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设计合成路线很有趣，希望大家课后多多练习！